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STUDENT SKILLS DEVELOPMENT MANAGEMENT SYSTEM

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA4305- Internet Programming

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We, M Ramya Sri (192372144) of the [CSE AI,CSE], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Student Skills Development management system” is the result of our own bonafide efforts.

To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 01.09.2026

Signature of the Students with Name

M Ramya Sri



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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Student Skills Development management system” has been carried out by Dr Marie Claude .N under the supervision of Dr. Sudhagar D the concerned faculty and is submitted in partial fulfilment of the requirements for the current semester of the Bachelor of Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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INTERNAL EXAMINER

EXTERNAL EXAMINER



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Individual Contribution Statement

Student	Responsibility	Design & Development	Testing	Report
T Gnaneswari (192372183)	Profile & Skills	Login, profile, skill module	Module testing	Intro duction 25%
Madhavi (192372133)	Skill Assessment	Assessment & score calculation	Assessment testing	Imple mentat ion 25%
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				Total 100%

ABSTRACT

The Student Skills Development Management System is a web-based platform developed to help students in identifying, developing, assessing, and monitoring their technical and soft skills. In today's competitive academic and employment environment, students are required to demonstrate not only subject knowledge but also programming, communication, problem-solving, teamwork, leadership, and other career-oriented skills. However, conventional academic management platforms generally concentrate on marks, attendance, and course completion while offering limited facilities for continuous monitoring of skill development.

The system offers a centralized environment in which students can create their profiles, select career goals, add existing skills, take skill assessments, monitor their performance, identify skill gaps, access learning recommendations, and track their development over time.

Skills are categorized into technical and soft skills and can be represented using proficiency levels and progress percentages.

One of the key features of the system is Skill Gap Analysis, that compares the student's existing proficiency with the level required for a selected career goal. Using the gaps identified, the system recommends areas that require improvement. Students are able to participate in learning activities and assessments, update their progress, and unlock achievements as they develop their skills.

An Admin/Faculty Dashboard is also provided so that authorized faculty members to monitor students, view assessment results, analyze skill development, identify students who require improvement, and observe department-level performance through charts and reports.

The intended outcome is an accessible and user-friendly platform that improves transparency in student skill development and helps students prepare for academic, professional, and placement opportunities.

Keywords: Student Skill Development, Skill Assessment, Skill Gap Analysis, Career Development, Recommendation System, Progress Tracking, Web Application, Student Management.

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CHAPTER 1: INTRODUCTION

1.1 Background Information

The development of appropriate technical and professional skills has become an important part of higher education. Academic marks provide information about a student's performance in a particular subject, but they may not completely represent the student's practical knowledge, technical abilities, communication skills, problem-solving ability, teamwork, leadership, and career readiness.

Students pursuing engineering and technology programs are required to continuously develop skills according to changing industry requirements. For example, a student interested in becoming an Artificial Intelligence Engineer may require strong knowledge of Python, Machine Learning, Data Structures, SQL, mathematics, communication, and problem solving.

Similarly, a student interested in becoming a Full Stack Developer may require HTML, CSS, JavaScript, databases, backend development, Git, and deployment knowledge.

Students often use multiple websites, learning platforms, certificates, assessments, and personal records to maintain information about their skills. Because this information is distributed across different platforms, monitoring overall development becomes difficult.

Faculty members also face challenges in identifying the strengths and weaknesses of individual students. Academic performance alone may not provide sufficient information about whether a student possesses the skills required for internships, placements, projects, or higher studies.

The Student Skills Development Management System is proposed as a centralized platform for maintaining and analyzing student skills. The platform enables students to create profiles, select career goals, maintain technical and soft skills, complete assessments, identify skill gaps, receive recommendations, track progress, and view achievements.

The platform has been designed to answer practical questions such as:

- What are my strongest skills?

- Which skills need improvement?
- What is my current proficiency level?
- Which skills are required for my career goal?
- What is the gap between my current and target skill levels?
- Which skill should I improve first?
- How has my skill performance changed over time?
- Am I becoming ready for my selected career path?

1.2 Project Objectives

The primary objective of the Student Skills Development Management System is to develop an integrated web-based platform for managing, assessing, analyzing, and improving student skills.

Specific objectives

1. To develop a secure student registration and authentication system.
2. To create and maintain individual student profiles.
3. To allow students to record technical and soft skills.
4. To classify student skills according to proficiency levels such as Beginner, Intermediate, Advanced, and Expert.
5. To conduct skill-based assessments for evaluating student knowledge.
6. To automatically calculate assessment scores and performance percentages.
7. To identify skill gaps by comparing current student skills with required skills.
8. To provide recommendations for improving weaker skills.
9. To provide learning resources for selected skills.
10. To track student skill development over time.

11. To provide graphical representations of student performance.

12. To introduce achievements and badges for encouraging continuous skill development.

13. To generate student skill-development reports.

14. To provide an Admin/Faculty Dashboard for monitoring student development.

15. To support better preparation for internships, placements, projects, and career opportunities.

1.3 Significance of the Project

The significance of the Student Skills Development Management System lies in the fact that it extends student management beyond academic marks and attendance.

Centralized Skill Management

Students are able to maintain their technical and soft skills within a single platform instead of maintaining separate records.

Skill Awareness

The system helps students understand their strengths and identify skills that require improvement.

Career-Oriented Development

Students are able to select a career goal and compare their current skill levels with the skills expected for that career.

Continuous Assessment

Assessments provide a measurable method of evaluating student knowledge instead of relying completely on self-reported proficiency.

Skill Gap Identification

The system automatically determines the difference between current and target skill levels.

Personalized Improvement

Students receive recommendations based on their identified skill gaps.

Faculty Monitoring

Faculty members can monitor individual and overall student development through the administrative dashboard.

Data Visualization

Charts and progress indicators make skill development easier to understand.

1.4 Scope

The functional scope of the Student Skills Development Management System includes: • Student registration and login.

- Faculty/Admin authentication.
- Student profile creation.
- Career goal selection.
- Technical skill management.
- Soft skill management.
- Skill proficiency tracking and Skill assessments.
- Automatic score calculation.
- Skill gap analysis and Skill recommendations.
- Learning-resource management.
- Course progress tracking.
- Student progress visualization.
- Achievement and badge management.

- Student skill reports and Admin student monitoring.
- Department-wise analytics.
- Performance visualization.

Limitations / Out of Scope The present version does not guarantee employment or placement opportunities. Skill scores generated by the application represent performance within the available assessments and should not be considered a replacement for formal professional certification.

1.5 Methodology Overview

The development methodology is organized into the following stages.

Step 1 – Student Registration A student begins by creating an account and providing personal and academic information.

Step 2 – Profile Creation Student details such as name, register number, department, year, semester, interests, and career goal.

Step 3 – Skill Selection Students enter their existing technical and soft skills.

Step 4 – Skill Assessment Students attempt assessments associated with selected skills.

Step 5 – Score Calculation The application determines the number of correct responses, total score, percentage, and corresponding skill level.

Step 6 – Skill Gap Analysis The student's present performance is compared against the target skill level required for the selected career.

Step 7 – Recommendation The application identifies and recommends skills that need additional development.

Step 8 – Learning and Development Students use the available learning resources and record their learning progress.

Step 9 – Faculty Monitoring Faculty members review student skill performance through the Admin Dashboard.

Step 10 – Report Generation A consolidated report of skill development is produced by the system.

CHAPTER 2: PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Students encounter several difficulties when trying to develop their skills in a systematic manner for academic and professional growth.

The first major issue is the lack of centralized skill tracking. Students may learn different skills through academic courses, online learning platforms, internships, workshops, projects, and certifications. However, this information is often maintained separately.

Another difficulty is difficulty in identifying weaknesses. Students may know multiple technologies but may not have a clear understanding of their actual proficiency levels.

A further challenge is career-skill mismatch. A student may have several skills but may not know whether those skills are sufficient for a selected career goal.

Another concern is limited continuous assessment. Students often evaluate themselves informally without using structured assessments.

An additional issue is lack of progress monitoring. Without historical performance data, it becomes difficult to understand whether a student's skill level is actually improving.

A further administrative challenge is faculty monitoring. Faculty members may have difficulty individually monitoring the skill-development progress of a large number of students.

The final challenge is fragmented learning resources. Students frequently use different platforms to learn individual skills without a structured development path.

The proposed system responds to these issues through centralized skill management, assessments, skill-gap analysis, recommendations, learning tracking, analytics, and faculty monitoring.

2.2 Evidence of the Problem

Current employment opportunities increasingly expect candidates to demonstrate practical and professional skills in addition to academic qualifications. Technical roles may require programming, database management, problem solving, cloud technologies, artificial intelligence, data analysis, web development, or cybersecurity skills.

Recruiters also evaluate communication, teamwork, adaptability, leadership, presentation ability, and critical thinking.

Conventional academic systems generally maintain information such as: • Student details • Attendance • Subject marks • Examination results • Course registration These systems may provide limited information about the continuous development of career- oriented skills.

This creates a need for a system that links: Student Profile Skills Career Goal Assessment Skill Gap Learning Progress

Existing Approaches and Limitations

Existing Approach / Limitation Traditional student portals Mainly focus on academic information Manual skill records Difficult to maintain and analyze Self-assessment May not accurately measure knowledge Individual learning platforms Skill information remains distributed Academic marks Do not completely represent practical skills Manual faculty monitoring Difficult for large numbers of students Generic learning recommendations May not consider individual skill gaps

2.3 Stakeholders

Stakeholder Interest / Requirement Students Identify and improve skills Faculty Monitor student development Mentors Guide students based on weaknesses Placement Team Understand student career readiness Department Analyze overall skill performance Administrators Manage users and system information Recruiters Potentially understand candidate capabilities Institution Improve student skill-development activities

2.4 Supporting Data / Research

The application depends on structured information covering students, skills, assessments, career requirements, courses, and progress.

A student profile may include:

- Student name • Register number • Department
- Year • Semester • Email • Interests • Career goal

Skill records may contain:

- Skill name • Skill category • Current level • Target level • Progress percentage • Assessment score
- Status

Career-skill mappings may contain:

- Career name • Required skills • Target proficiency • Skill priority

Assessment records may include:

- Skill • Question • Options • Correct answer • Difficulty

CHAPTER 3: SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The project uses an iterative software-development process organized into a sequence of phases.

Phase 1 – Requirement Analysis Requirements from both students and faculty are first identified.

Phase 2 – Data Design Data structures for students, skills, assessments, courses, achievements, and progress are then designed.

Phase 3 – System Design The architecture, modules, database collections, interfaces, and workflows are planned during this phase.

Phase 4 – Frontend Development The frontend is implemented for registration, login, dashboard, skills, assessments, recommendations, progress, and administration.

Phase 5 – Backend Development Backend APIs are developed to handle authentication, skill management, assessment processing, progress tracking, and administration.

Phase 6 – Database Development Database collections are established for users, skills, assessments, results, courses, achievements, and progress.

Phase 7 – Skill Analysis Development Algorithms for calculating scores and analyzing skill gaps are incorporated.

Phase 8 – Integration The frontend, backend, database, and analytics layers are brought together.

Phase 9 – Testing

Functional, validation, authentication, usability, and performance tests are carried out.

Phase 10 – Evaluation The completed application is assessed using functionality, calculation accuracy, usability, and response time.

3.2 Tools and Technologies Used

Category Technology Programming Language Python Frontend HTML, CSS, JavaScript UI Framework Bootstrap Backend FastAPI Database MongoDB Visualization Chart.js API Architecture REST API

Server Uvicorn Development Environment VS Code / Gravity Browser Google Chrome Version Control Git / GitHub Python and FastAPI are appropriate choices for backend services because they provide a structured approach for implementing APIs and application logic.

MongoDB offers flexible document-oriented storage for student profiles, skills, assessments, and activity information.

Chart.js is utilized to visualize student skill development and assessment performance graphically.

3.3 Solution Overview

The application is organized into several major functional components.

1. User Authentication Module

Handles registration, login, logout, and role-based access securely.

2. Student Profile Module

The module maintains:

- Name • Register Number • Email • Department • Year • Semester • Interests • Career Goal

3. Skill Management Module

Enables students to create, edit, update, and delete technical and soft skills.

4. Assessment Module

Runs skill-focused quizzes and calculates the results automatically.

5. Skill Gap Analysis Module

Evaluates the student's current proficiency against the defined target.

6. Recommendation Module

Recommends improvement areas based on the detected gaps.

7. Learning Module

Offers learning resources and lets students record completion.

8. Progress Module

Shows skill and assessment progress using charts.

9. Achievement Module

Awards badges when defined milestones are achieved.

10. Admin/Faculty Module

Enables authorized faculty to monitor student development.

3.4 System Architecture

The overall architecture follows this flow:

The architecture keeps the presentation, application, processing, and database layers separate, making the system easier to maintain and extend.

3.5 Functional Modules

Module 1: User Authentication Students and faculty members securely access the application using authorized credentials.

Module 2: Student Profile Stores personal, academic, career, and interest information.

Module 3: Skill Management Students maintain technical and soft skills.

Module 4: Skill Assessment Students are evaluated through questions associated with particular skills.

Module 5: Score Calculation Submitted answers are assessed and converted into percentage values.

Module 6: Skill Gap Analysis The application calculates the difference between current and required proficiency.

Module 7: Recommendation Skills showing larger gaps are assigned greater improvement priority.

Module 8: Learning Resources Students are able to access learning content for targeted skills.

Module 9: Progress Tracking Skill growth is displayed using progress bars and charts.

Module 10: Achievements Predefined milestones are recognized through badges.

Module 11: Admin Dashboard Faculty can review students, assessment outcomes, and progress.

Module 12: Report Generation The application produces a consolidated student skill-development report.

3.6 Algorithm and Methodology

Skill assessment procedure

START

Get Student ID

Select Skill

Load Assessment Questions

score = 0

FOR each question

Get student answer

IF student answer = correct answer

score = score + 1

END IF

END FOR

percentage = (score / total questions) × 100

IF percentage < 40

level = Beginner

ELSE IF percentage < 60

level = Intermediate

ELSE IF percentage < 80

level = Advanced

ELSE

level = Expert

END IF

Save result

Display score, percentage and level

STOP

Skill-gap analysis procedure

START

Get Student ID

Get Career Goal

Retrieve current skills Retrieve required skills for career FOR every required skill

current = Student Skill Percentage

required = Required Skill Percentage

gap = required - current

IF gap > 0

Status = Needs Improvement

ELSE

gap = 0

Status = Requirement Achieved

END IF

Store Result

END FOR

Sort skills according to largest gap

Generate recommendations

Display Skill Gap Analysis

STOP

Formula

Skill Gap = Required Skill Level – Current Skill Level

Example:

Required Python = 90% Current Python = 70% Skill Gap = $90 - 70 = 20\%$ Therefore, the student requires approximately 20% improvement in Python relative to the configured target.

3.7 Engineering Standards Applied

Standard / Principle Application

ISO/IEC 25010

Software quality evaluation ISO/IEC 27001 Principles Information security

IEEE 29148

Requirements engineering

IEEE 1016

Software design documentation Software Testing Principles Test documentation OWASP Principles Web application security The sample report similarly considers software quality, requirements engineering and web- security principles when discussing engineering standards.

ISO/IEC 25010

The application may be evaluated using the following criteria:

- Functional suitability • Performance efficiency • Usability • Reliability • Security • Maintainability
- Compatibility • Portability

Security Considerations

Authentication and authorization are intended to restrict unauthorized access to student information.

Input validation reduces the possibility of invalid data being submitted.

Passwords must not be stored directly as plain text.

Role-based permissions restrict students from administrative functions.

3.8 Solution Justification

The proposed solution is appropriate because conventional academic systems primarily provide academic information and may not provide detailed continuous skill-development analysis.

A marks-management application can show examination performance but may not identify career-specific skill gaps. An online learning platform may provide courses but may not maintain institution-level student development information. A self-assessment system can measure individual skills but may not provide continuous progress tracking.

The Student Skills Development Management System integrates these components: Student Profile → Skills → Assessment → Skill Gap → Recommendation → Learning → Progress → Analytics
This results in a more integrated platform for supporting student skill development.

CHAPTER 4: RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The performance of the Student Skills Development Management System can be assessed using several parameters.

Functional Accuracy The application is expected to correctly handle registration, authentication, skill management, assessment, and progress operations.

Assessment Accuracy Assessment answers should be correctly evaluated.

Skill Gap Accuracy The difference between target and current skill levels should be calculated correctly.

Recommendation Relevance Recommendations are required to match the student's actual identified skill gaps.

Data Accuracy Student profile, skill, assessment, and progress information should be correctly stored and retrieved.

Usability Users should be able to navigate the application and understand their skill-development information.

Performance Pages and API requests should respond within an acceptable time.

4.2 Output Parameters

Parameter	Expected Outcome
Registration	Successful account creation
Authentication	Secure user access
Student Profile	Correct information storage
Skill Management	Correct skill CRUD operations
Assessment	Accurate score generation
Skill Level	Correct proficiency classification
Skill Gap	Correct difference calculation
Recommendations	Relevant improvement suggestions
Progress	Correct graphical representation
Achievement	Badge unlocked at required milestone
Admin Analytics	Correct student summaries
Report	Accurate consolidated information

4.3 Test Cases and Expected/Actual Results

Test Case	Expected Result	Actual Result	Status
Student Registration	Account created	Account created	PASS

Valid Login	Dashboard displayed	Dashboard displayed	PASS
Invalid Login	Error message	Error displayed	PASS

Add Skill	Skill saved	Skill saved	PASS
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Edit Skill	Skill updated	Skill updated	PASS
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Delete Skill	Skill removed	Skill removed	PASS
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Assessment Score	generated	Score generated	PASS
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Skill Gap	Gap correctly calculated	Correct result	PASS
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Recommendation	Weak skills recommended	Recommendations shown	
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PASS

Course Progress Progress updated Progress updated

PASS

Achievement Badge unlocked Badge displayed

PASS

Admin Dashboard Student information displayed Information displayed

PASS

Important: Change the "Actual Result" and "PASS" values according to your real testing before submission.

Test Success Rate $\text{Test Success Rate (\%)} = \frac{\text{Passed Test Cases}}{\text{Total Test Cases}} \times 100$ If all 12 tests pass: $\text{Test Success Rate} = \frac{12}{12} \times 100 = 100\%$

4.4 Challenges Encountered

Challenge 1 – Handling Multiple Skill Categories Students are able to possess several technical and soft skills with different proficiency levels.

Solution: Skills are stored using structured categories, proficiency levels, and percentages.

Challenge 2 – Measuring Skill Proficiency Self-reported skill levels may not always represent actual knowledge.

Solution: Skill assessments are included to provide additional measurable evidence.

Challenge 3 – Linking Careers with Skills Different careers require different combinations of skills.

Solution: Career goals are mapped to predefined target skills and proficiency levels.

Challenge 4 – Making Recommendations Relevant Generic recommendations may not be useful to every student.

Solution: Recommendations are generated according to identified skill gaps.

Challenge 5 – Presenting Performance Data Large amounts of student performance information can be difficult to understand as plain text.

Solution: Charts and progress indicators are used.

Challenge 6 – Controlling Access Students should not be allowed to access administrative functionality.

Solution: Role-based authorization is implemented.

4.5 Possible Improvements

Future releases of the application may provide:

- 1. Machine-learning-based skill recommendations.**
- 2. AI-based career recommendations.**
- 3. Resume skill extraction.**
- 4. Automatic resume analysis.**
- 5. Placement-readiness prediction.**
- 6. Company-specific skill-gap analysis.**
- 7. Integration with online learning platforms.**
- 8. Certificate verification.**
- 9. Internship recommendation.**
- 10. Project recommendation according to student skills.**
- 11. AI-based mock interviews.**
- 12. Coding assessment integration.**
- 13. Mobile application support.**
- 14. Faculty mentoring and appointment features.**

15. Email and mobile notifications.

16. Advanced department analytics.

17. Peer skill comparison.

18. Personalized learning paths.

4.6 Recommendations

The following measures are recommended for future implementation: • Regularly update career-specific skill requirements.

- Maintain a larger assessment question bank.
- Validate assessment questions before publishing.
- Provide transparent skill scoring.
- Implement strong authentication and access control.
- Regularly back up student information.
- Conduct usability testing with actual students.
- Collect student feedback.
- Provide diverse learning resources.
- Continuously improve skill recommendations.

4.7 SDG RELEVANCE

SDG 4 – Quality Education The project is mainly aligned with Sustainable Development Goal 4: Quality Education.

The platform promotes continuous learning by enabling students to identify existing skills, detect weaknesses, complete assessments, and monitor development.

It can support students in developing technical and professional competencies in addition to academic knowledge.

SDG 8 – Decent Work and Economic Growth The project is also connected with SDG 8: Decent Work and Economic Growth.

Career-oriented skill development can improve students' preparation for internships, placements, entrepreneurship, and employment opportunities.

By identifying skill gaps before students enter the employment market, the platform can support better career preparation.

CHAPTER 5: REFLECTION ON LEARNING AND

PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

The project offered practical experience in applying concepts from computer science, database management, software engineering, web development, data visualization, algorithms, and application security.

The project demonstrated how theoretical concepts can be converted into a practical student-development platform.

Technical Skills

The development work strengthened technical knowledge in: • Python programming • FastAPI • HTML

•

CSS

- JavaScript • Bootstrap • MongoDB • REST API development • Authentication • Database operations • Chart.js • Data visualization • Software testing

- Git/GitHub

Problem Solving and Critical Thinking

The project required solving several connected problems.

For example, a student may have a high proficiency in one technical skill but a significant gap in another skill required for the selected career.

The system therefore requires analysis of multiple skill values rather than simply storing student information.

Working on the project improved the ability to break a large problem into smaller functional modules and integrate them into a complete application.

5.2 Challenges Encountered and Overcome

One difficulty involved creating a flexible structure capable of maintaining different skills for different students.

Another difficulty was developing a method for converting assessment performance into understandable skill levels.

Skill-gap analysis additionally required the current student level and target career level to be represented consistently.

Integrating the frontend with the backend required careful management of API requests, responses, validation, and database operations.

Authentication and authorization required separation between student and administrative access.

Addressing these challenges strengthened debugging, logical thinking, application design, and testing skills.

5.3 Application of Engineering

Software-engineering principles offered a systematic approach to the development of the project. Requirement analysis helped identify student and faculty requirements. Modular design helped divide the system into authentication, profile, skill, assessment, recommendation, progress, and administrative components. Database-design principles were used while creating structures for users, skills, assessments, and results.

5.4 Insights into the Industry

The project demonstrated that modern software development extends beyond programming alone.

A complete software product involves:

- Requirement analysis • User interface design • Backend development • Database management • Security • Testing • Performance evaluation • Deployment • Documentation • Maintenance • Continuous improvement

5.5 Conclusion of Personal Development

The project contributed to both technical growth and professional development.

From a technical perspective, the project strengthened understanding of frontend development, backend APIs, databases, authentication, algorithms, analytics, and testing.

The development process further improved problem-solving, communication, teamwork, documentation, debugging, and time-management skills.

The work showed how software-engineering concepts can be applied to solve a practical educational problem.

CHAPTER 6: CONCLUSION

The Student Skills Development Management System was created as a centralized platform for identifying, assessing, managing, and improving student skills.

The project extends beyond the limitations of conventional academic management systems by extending student monitoring beyond marks and attendance. The proposed platform maintains technical and soft skills, performs assessments, calculates proficiency levels, identifies skill gaps, provides recommendations, monitors learning progress, and displays development through analytics.

A key strength of the system is its integrated design. Instead of maintaining student skills, assessments, learning activities, and progress separately, these components are connected within a single platform.

The Skill Gap Analysis module enables students to see the difference between their current capabilities and career-oriented target requirements. The recommendation component then guides students regarding which skills require additional attention.

The Admin/Faculty Dashboard adds further value of the system by allowing faculty members to monitor student performance and development.

The project shows the practical use of web development, database management, software engineering, algorithms, data visualization, testing, and security principles to an educational problem.

Future versions of the system can incorporate machine learning, AI-based career guidance, resume analysis, placement-readiness prediction, automated certificate verification, learning- platform integration, and mobile application support.

Overall, the system illustrates how digital technology can support continuous learning, career preparation, skill awareness, and student development.

REFERENCES

For the final submission, include only references that you actually used in the project.

Suitable categories are:

1. **FastAPI – Official Documentation.**
2. **MongoDB – Official Documentation.**
3. **Python – Official Documentation.**
4. **Bootstrap – Official Documentation.**
5. **Chart.js – Official Documentation.**
6. **OWASP Foundation – Web Application Security Guidelines.**
7. **ISO/IEC 25010 – Systems and Software Quality Models.**
8. **ISO/IEC/IEEE 29148 – Requirements Engineering.**
9. Research papers related to student skill assessment and career recommendation systems.
10. Relevant educational skill-development datasets, if any dataset is actually used.

APPENDICES

Appendix A – Sample Student Input
Input Example Name Student A Register Number 192372183
Department

CSE-AI

Year

IV

Semester

VII

Career Goal AI Engineer Python 70% Machine Learning 55%

SQL

65% Communication 80% Appendix B – Sample System Output Career Goal: AI Engineer Skill

Current Required Gap Result Python 70% 90% 20% Improve Machine Learning 55% 85% 30%
Improve

SQL

65% 70% 5% Improve Communication 80% 70% 0% Achieved Recommendation Priority 1: Improve
Machine Learning – Gap 30% Priority 2: Improve Python – Gap 20% Priority 3: Improve SQL – Gap
5% Strength: Communication – Required level achieved.

Appendix C – Execution Screenshots

After completing the website, paste your actual screenshots in this order:

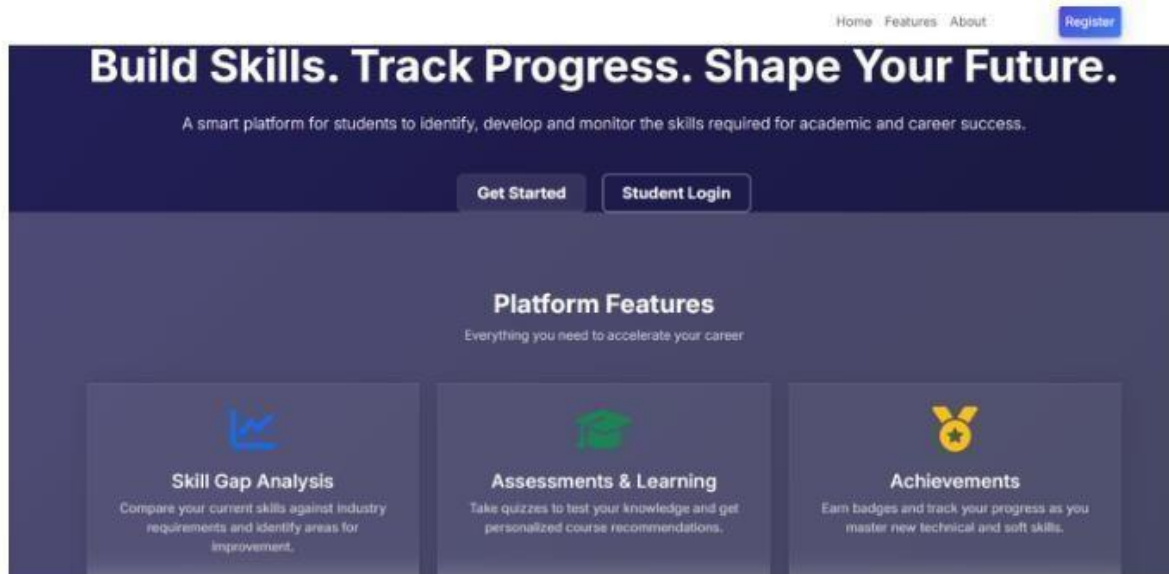


Figure 1 – Home Page

The screenshot shows the student registration page. It has a navigation bar with 'Home', 'Features', 'About', and a 'Register' button. The main heading is 'Student Registration' with the subtext 'Create your account to start tracking skills'. The registration form includes fields for 'Full Name' (filled with 'ganeswari'), 'Register Number' (filled with '192372183'), 'Email address' (filled with 'ganeswari2183.sse@saveetha.com'), 'Department' (filled with 'CSE-AI'), 'Year' (filled with '4'), 'Semester' (filled with '07'), 'Password', and 'Confirm Password'. There is a 'Register Account' button and a link for 'Already have an account? Login here'.

Figure 2 – Student Registration Page

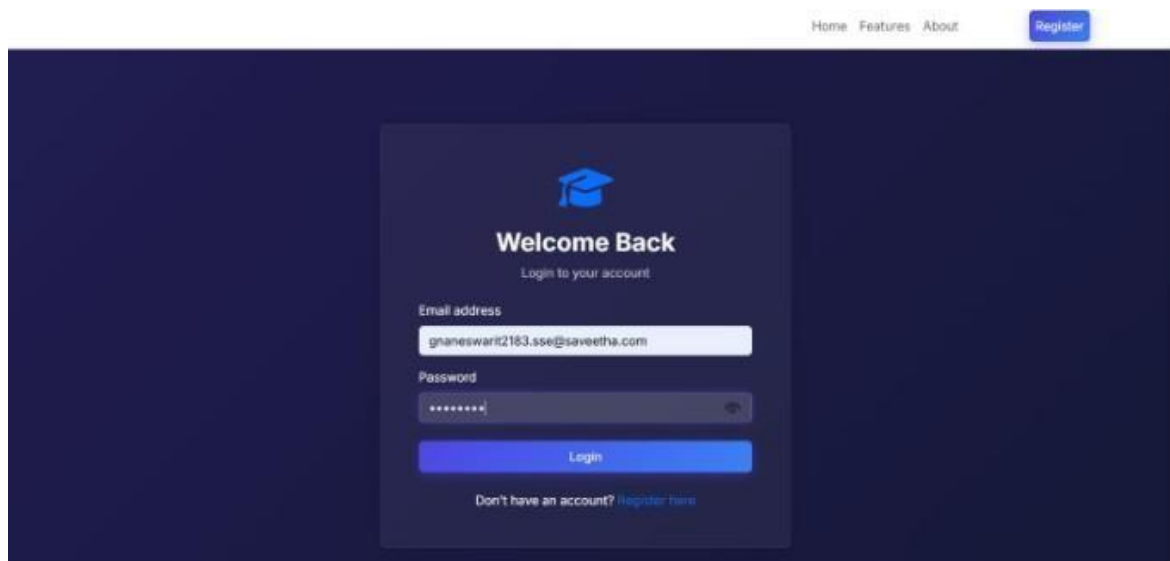


Figure 3 – Student Login Page



Figure 4 – Student Dashboard

Student Profile

kusuma
CSE-AI (Year 4)
192372211

Email (Non-editable): ikusumareddy23@gmail.com
Department: CSE-AI
Year: 4
Semester: 7

Career & Interests

Career Goal: Software Developer
Interests (Comma separated): Coding, Technology

Profile Image Update
Choose File: No file chosen

Save Changes

Figure 5 – Student Profile

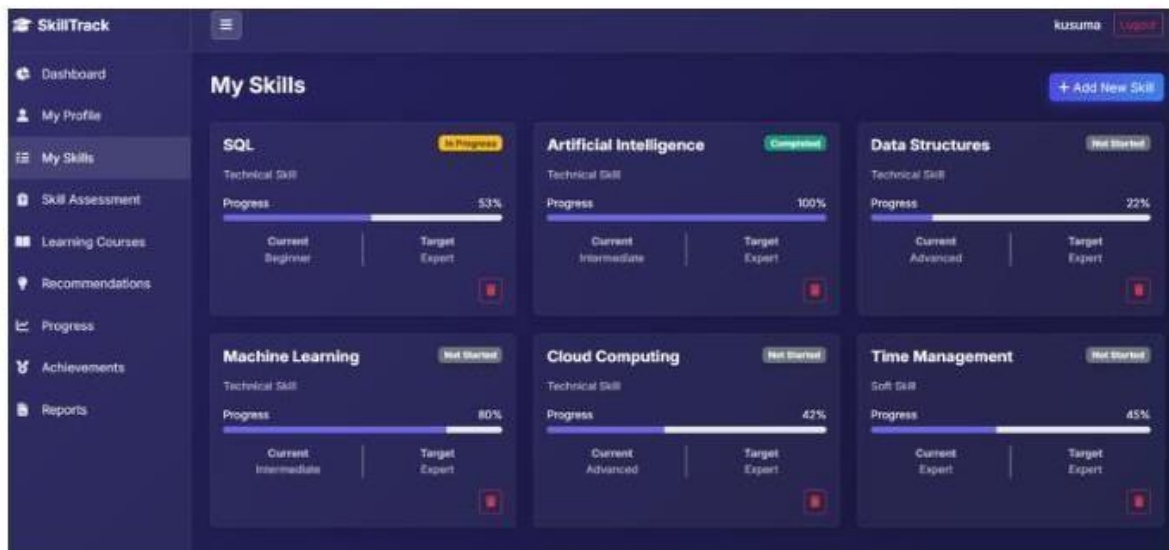


Figure 6 – My Skills

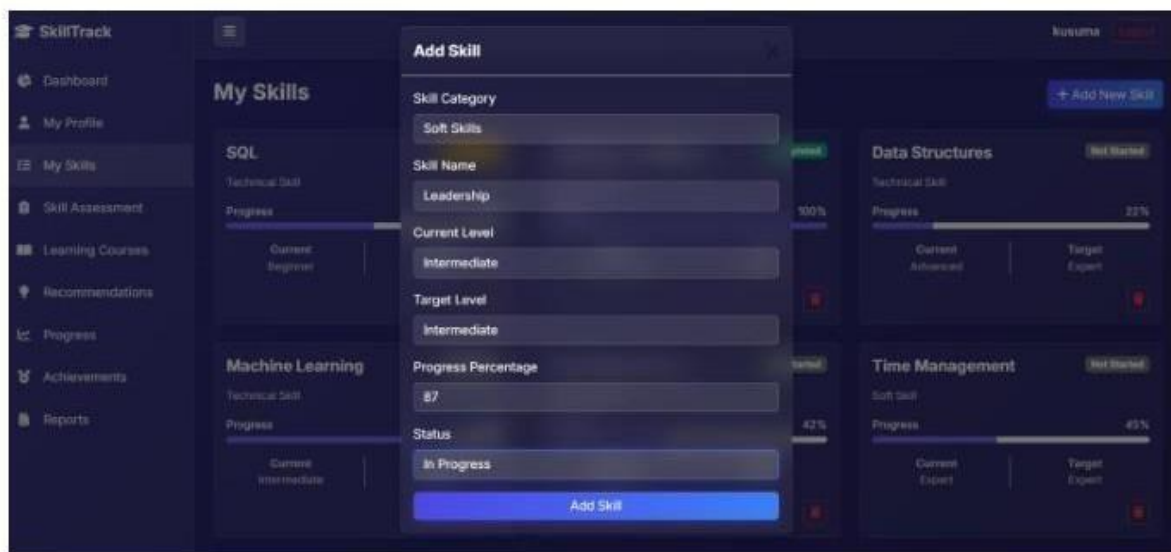


Figure 7 – Add/Update Skill

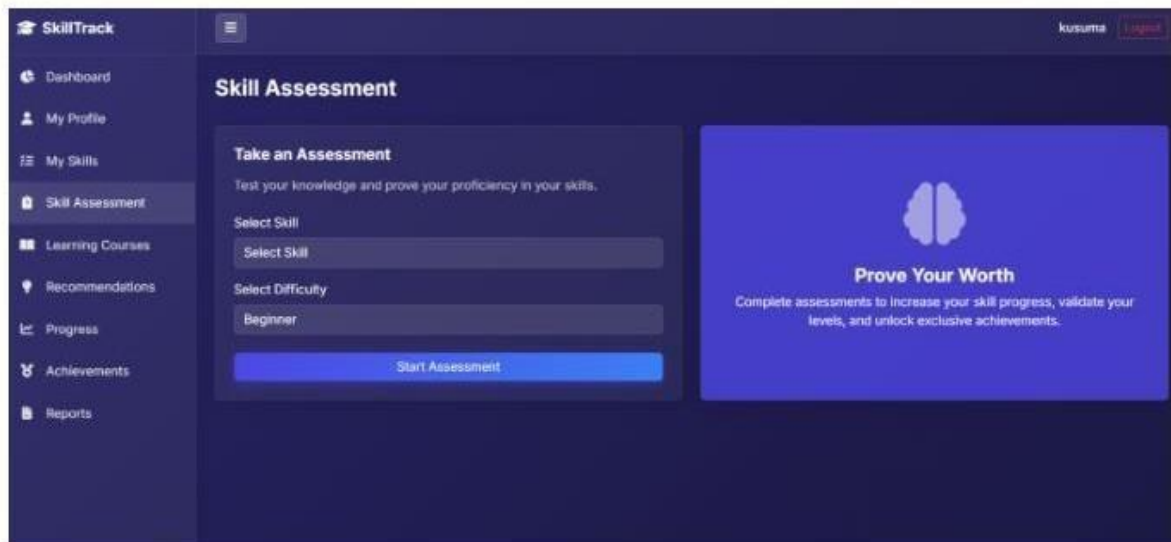


Figure 8 – Skill Assessment

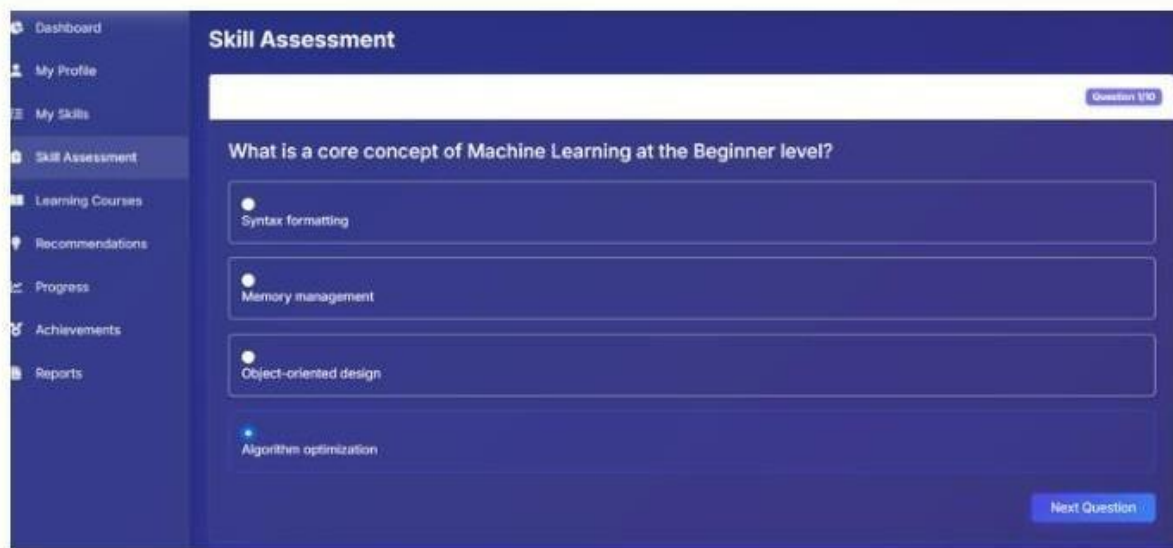


Figure 9 – Assessment Result

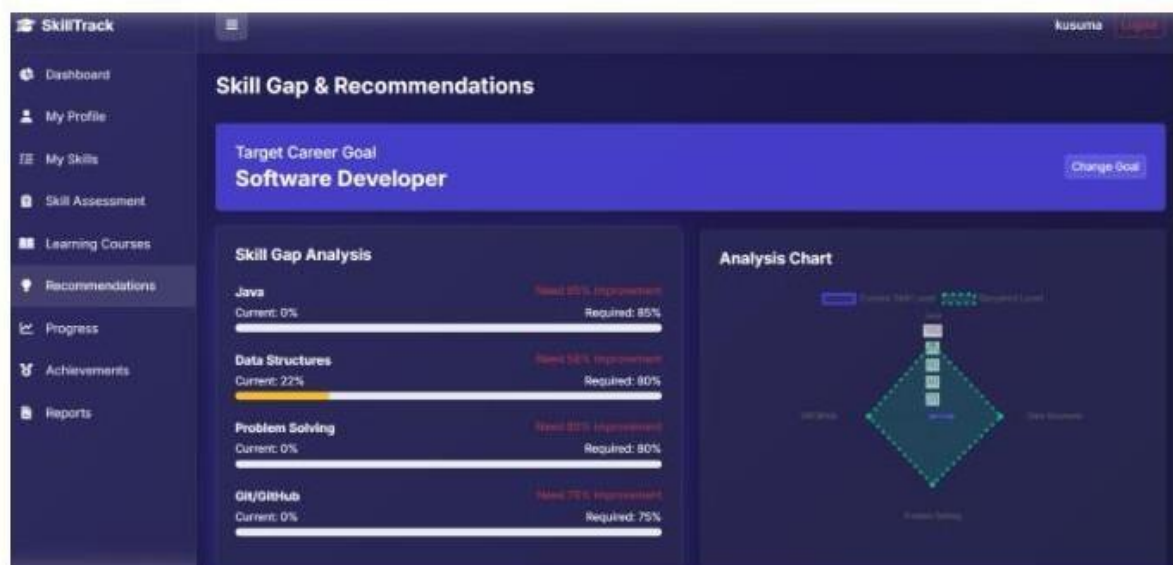


Figure 10 – Skill Gap Analysis

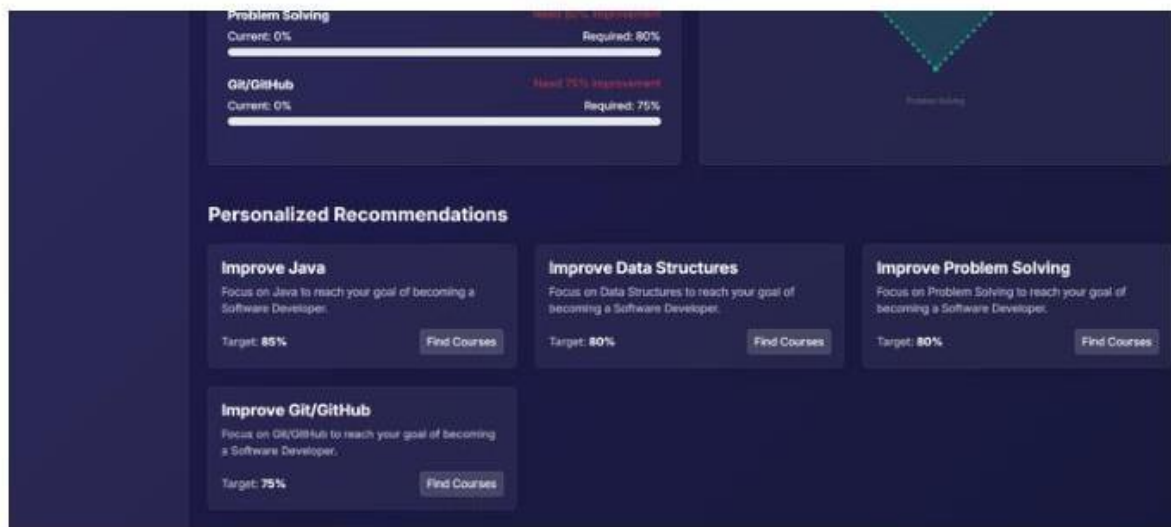


Figure 11 – Personalized Recommendations

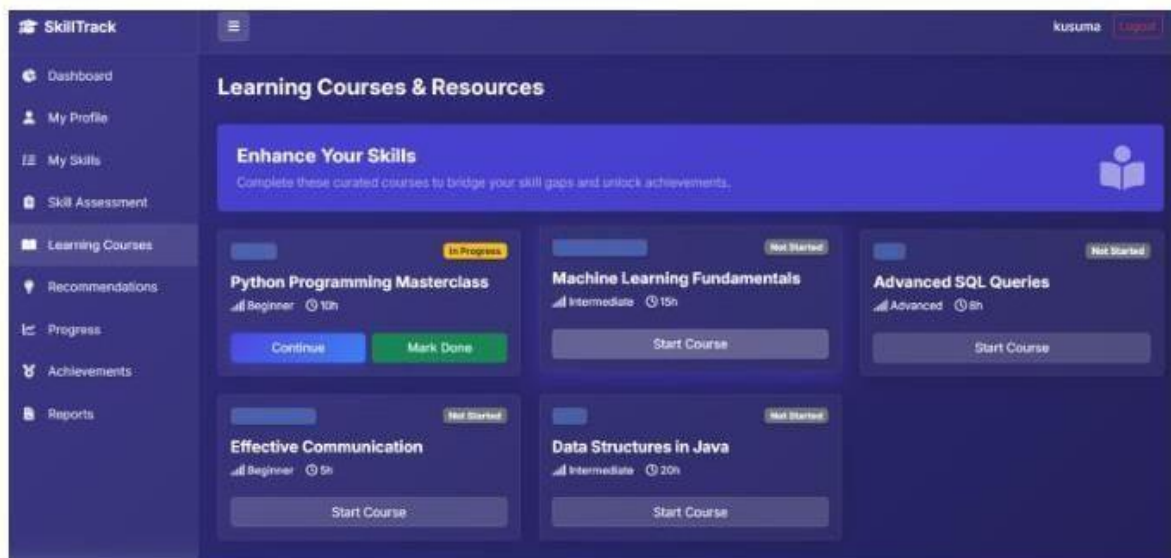


Figure 12 – Learning Courses

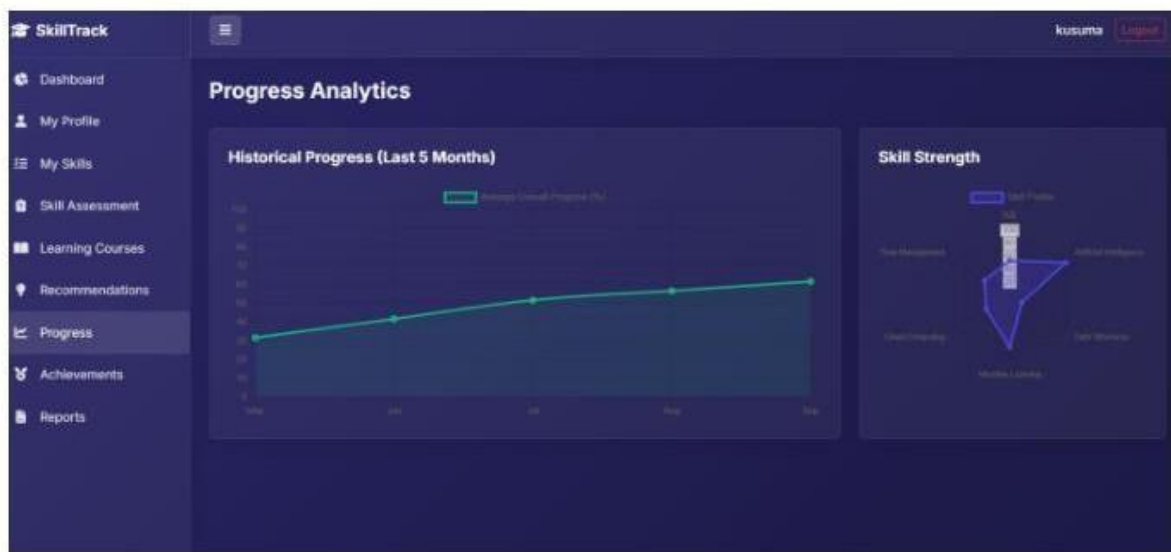


Figure 13 – Progress Analytics

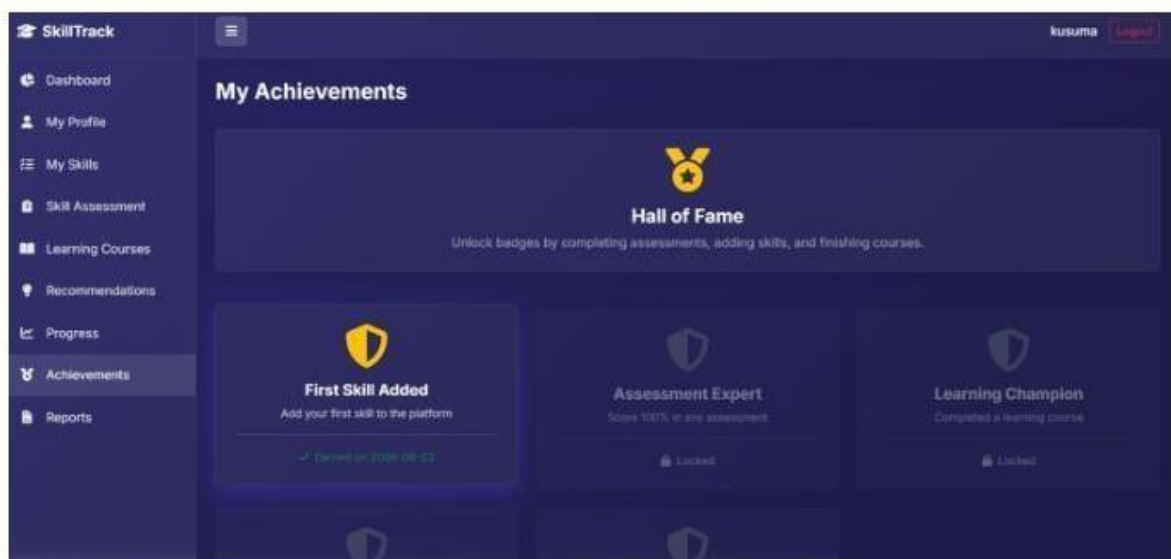


Figure 14 – Achievements



Figure 15 – Student Skill Report



Figure 16 – Admin Dashboard

Student Directory						All Departments
Name	Reg Number	Department	Year	Skills Tracked	Assessments	
Student 1	REG0000001	CS001	3	14	10	
Student 2	REG0000002	CS001	3	10	8	
Student 3	REG0000003	CS001	3	8	5	
Student 4	REG0000004	CS001	3	15	12	
Student 5	REG0000005	CS001	3	12	9	
Student 6	REG0000006	CS001	3	10	8	
Student 7	REG0000007	CS001	3	10	8	
Student 8	REG0000008	CS001	3	10	8	
Student 9	REG0000009	CS001	3	10	8	
Student 10	REG0000010	CS001	3	10	8	
Student 11	REG0000011	CS001	3	10	8	
Student 12	REG0000012	CS001	3	10	8	
Student 13	REG0000013	CS001	3	10	8	
Student 14	REG0000014	CS001	3	10	8	
Student 15	REG0000015	CS001	3	10	8	
Student 16	REG0000016	CS001	3	10	8	
Student 17	REG0000017	CS001	3	10	8	
Student 18	REG0000018	CS001	3	10	8	
Student 19	REG0000019	CS001	3	10	8	
Student 20	REG0000020	CS001	3	10	8	
Student 21	REG0000021	CS001	3	10	8	
Student 22	REG0000022	CS001	3	10	8	
Student 23	REG0000023	CS001	3	10	8	
Student 24	REG0000024	CS001	3	10	8	
Student 25	REG0000025	CS001	3	10	8	
Student 26	REG0000026	CS001	3	10	8	
Student 27	REG0000027	CS001	3	10	8	
Student 28	REG0000028	CS001	3	10	8	
Student 29	REG0000029	CS001	3	10	8	
Student 30	REG0000030	CS001	3	10	8	
Student 31	REG0000031	CS001	3	10	8	
Student 32	REG0000032	CS001	3	10	8	
Student 33	REG0000033	CS001	3	10	8	
Student 34	REG0000034	CS001	3	10	8	
Student 35	REG0000035	CS001	3	10	8	
Student 36	REG0000036	CS001	3	10	8	
Student 37	REG0000037	CS001	3	10	8	
Student 38	REG0000038	CS001	3	10	8	
Student 39	REG0000039	CS001	3	10	8	
Student 40	REG0000040	CS001	3	10	8	
Student 41	REG0000041	CS001	3	10	8	
Student 42	REG0000042	CS001	3	10	8	
Student 43	REG0000043	CS001	3	10	8	
Student 44	REG0000044	CS001	3	10	8	
Student 45	REG0000045	CS001	3	10	8	
Student 46	REG0000046	CS001	3	10	8	
Student 47	REG0000047	CS001	3	10	8	
Student 48	REG0000048	CS001	3	10	8	
Student 49	REG0000049	CS001	3	10	8	
Student 50	REG0000050	CS001	3	10	8	
Student 51	REG0000051	CS001	3	10	8	
Student 52	REG0000052	CS001	3	10	8	
Student 53	REG0000053	CS001	3	10	8	
Student 54	REG0000054	CS001	3	10	8	
Student 55	REG0000055	CS001	3	10	8	
Student 56	REG0000056	CS001	3	10	8	
Student 57	REG0000057	CS001	3	10	8	
Student 58	REG0000058	CS001	3	10	8	
Student 59	REG0000059	CS001	3	10	8	
Student 60	REG0000060	CS001	3	10	8	
Student 61	REG0000061	CS001	3	10	8	
Student 62	REG0000062	CS001	3	10	8	
Student 63	REG0000063	CS001	3	10	8	
Student 64	REG0000064	CS001	3	10	8	
Student 65	REG0000065	CS001	3	10	8	
Student 66	REG0000066	CS001	3	10	8	
Student 67	REG0000067	CS001	3	10	8	
Student 68	REG0000068	CS001	3	10	8	
Student 69	REG0000069	CS001	3	10	8	
Student 70	REG0000070	CS001	3	10	8	
Student 71	REG0000071	CS001	3	10	8	
Student 72	REG0000072	CS001	3	10	8	
Student 73	REG0000073	CS001	3	10	8	
Student 74	REG0000074	CS001	3	10	8	
Student 75	REG0000075	CS001	3	10	8	
Student 76	REG0000076	CS001	3	10	8	
Student 77	REG0000077	CS001	3	10	8	
Student 78	REG0000078	CS001	3	10	8	
Student 79	REG0000079	CS001	3	10	8	
Student 80	REG0000080	CS001	3	10	8	
Student 81	REG0000081	CS001	3	10	8	
Student 82	REG0000082	CS001	3	10	8	
Student 83	REG0000083	CS001	3	10	8	
Student 84	REG0000084	CS001	3	10	8	
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Student 87	REG0000087	CS001	3	10	8	
Student 88	REG0000088	CS001	3	10	8	
Student 89	REG0000089	CS001	3	10	8	
Student 90	REG0000090	CS001	3	10	8	
Student 91	REG0000091	CS001	3	10	8	
Student 92	REG0000092	CS001	3	10	8	
Student 93	REG0000093	CS001	3	10	8	
Student 94	REG0000094	CS001	3	10	8	
Student 95	REG0000095	CS001	3	10	8	
Student 96	REG0000096	CS001	3	10	8	
Student 97	REG0000097	CS001	3	10	8	
Student 98	REG0000098	CS001	3	10	8	
Student 99	REG0000099	CS001	3	10	8	
Student 100	REG0000100	CS001	3	10	8	

Figure 17 – Student Management



Figure 18 – Admin Analytics